

RESEARCH AND DESIGN AUTOMATIC RAILWAY CROSSING WARNING SYSTEM BASED ON APPLICATION OF RADIO SOLUTION AND SATELLITE POSITIONING TECHNOLOGY

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Abstract

Railway accidents have been causing serious consequences, especially at RC - the intersection between railways and roads. Currently, the automatic RC warning system applied in the country is mainly imported from abroad. Due to the large investment costs, the lack of mastery of the technology and the limited functionality of the system, the investment in installation is still very modest. Therefore, there are many other RC that are in the state of "3 nos" (no barriers, no guards and no warnings signals), always potentially dangerous risks of traffic accidents at any time for people and vehicles to cross those RC. This paper will propose a new solution, design a automatic RC warning system based on the application of advanced technologies, in order to improve the system's capabilities and reduce investment costs as well as mastering the technology. Contribute to reducing traffic accidents and throughput capacity of the railway industry in Vietnam.

Keywords: Railway crossing, Radio, GPS, Control, Measurement, Warning, Fail safe.

Abbreviations

ARCWS	Automatic Railway Crossing Warning System
BTM	Balise Transmission Module
GLONASS	Globalnaya Navigatsionnaya Sputnikovaya Sistema
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GSM	Global System Mobile
HMI	Human Machine Interface
PTC	Positive Train Control
RC	Railway Crossing
RF	Radio Frequency

The railway industry used a magnetic sensor solution from an Austrian company. However, after a period of use, the Austrian system had to be dismantled because Vietnam's conditions operation were quite complicated: many types of vehicles cross the RC (from motor vehicles such as cars, agricultural vehicles, motorbikes to rudimentary vehicles such as bicycles, bullock carts...) generate interference, lead to false alarm signal. Another simpler solution, based on the principle of "rail circuit", was also used, this solution is based on the principle of placing contacts at the point of junction between two rails, when the train passes, the train wheel will connect those two contacts, a short-circuit signal is sent to the alarm system. This system has been installed at nearly 100 RC, but only 50% efficiency is achieved because Vietnam's climate is hot and humid, so the contacts are quickly oxidized, contact becomes less effective, so the operation becomes low effective. Optical solutions have also been tested, but do not seem to have completely solved the environmental problems, interference and impacts caused by people living in the surrounding area.

Up to now, the railway industry in Vietnam mainly uses seismic sensors and magnetic field sensors to detect train crossings. The system is imported from abroad, so the investment cost is quite high and the construction and installation of the system as well as maintenance and repair is complicated, so the number of automatic warning horizontal lines has been installed at a limited level. This leads to huge labor costs at guarded RC, and at the same time, many other RC are in a state of "3 no", causing traffic insecurity on these crossings lurking at any time. According to statistics, more than 80% of accidents are determined by objective factors, occurring at the self-opening aisles along the railway.

1. Introduction

Traffic safety in general and railway traffic safety in particular is a special concern of authorities and people because the number of accidents occurring every year is still at a worrying figure. Especially, the number of catastrophic accidents that kill and injure many people tends to increase. According to the analysis of experts, most of the railway traffic accidents occur at the intersection of the road with the railway.

There are 1,427 legal railway crossing on the Vietnam railways network. The average density of 2km of the main railway line has 1 railway crossing (according to the report of the Vietnam Railway Authority updated date: 26/11/2015). The specific data is summarized as follows:

Total: 1427 railways crossing.

There is guard: 503 railways crossing.

Automatic alarm signal: 217 railways crossing.

Level 3 has a sign: 707 railways crossing.

People's roads: 4042 railways crossing.



Fig. 1 The RC always potential the risk of traffic accidents

Currently, satellite navigation technology has developed to improve accuracy and reliability, which has been proven through theory and practice. The number of satellites is covered – anywhere on the earth's surface can receive satellite waves, and today's satellite data receiver module can receive many satellites from many company's satellites at the same time provide such as GPS, Galileo, GLONASS, Beidou with high frequency. In the country, the automatic railway crossing warning system applying radio solutions and GPS technology currently has no published research or application products, which only applies individually in certain applications. Some countries around the world have researched to apply this solution to RC, although their products have only been developed basically, but have brought high efficiency. However due to infrastructure floors, operating conditions of Vietnam's railway industry that foreign equipment can hardly apply. At the same time, the investment costs are high, the equipment operation is difficult because of not mastering the technology for foreign systems, so it has not been invested yet.

Thus, it is important and urgent to research design an automatic railway crossing warning system using radio solutions and GPS technology. At the same time,

research on adding a number of limited functions from foreign systems to further perfect the product, install additional railway crossing without automatic warning system as well as gradually replace systems the import system, aims to:

- + Reduce the large number of workers managing, exploiting and operating the system on railways crossing.
- + Reduce the cost and time to build an automatic RC warning system (the system does not need to dig roads, take cables like the current system, limit negative impacts from outside on the system, ...)
- + Due to mastering technology, it is possible to quickly and easily maintain, repair and replace when the system has problems.
- + The investment cost to build the system is low, the system is compact and easy to install, so the deployment of the system application can be replicated in many places, including the RC with little vehicle, in order to minimize the disaster traffic accidents, improve quality and operating capacity.

2. Evaluate the reliable working ability of the applied technology

The principle of safety, availability and reliability of the system is especially important for the railway industry, especially for automatic control systems in railways. The main technologies applied to the author's research direction are GPS technology and radio communication solutions, which are new technologies in application to the railway industry. Domestically, the application of this technology is still limited because of concerns about the system's reliability, however, it has been proven in the world and is increasingly widely applied.

In the world, the application of radio solutions and GPS technology has had many practical applications. These are solutions that bring high efficiency in railway management and operation to ensure safety and economic efficiency. In 2008, the U.S. Congress enacted an improved law to improve the entire railroad, in which the most significant change was the mandate that PTC active train control be integrated on most rail networks United States in 2015[1][2][3]. Train location is a basic function of PTC combined with wireless data link and is GPS-based [4]. Its basic principle is drawn on figure 2.

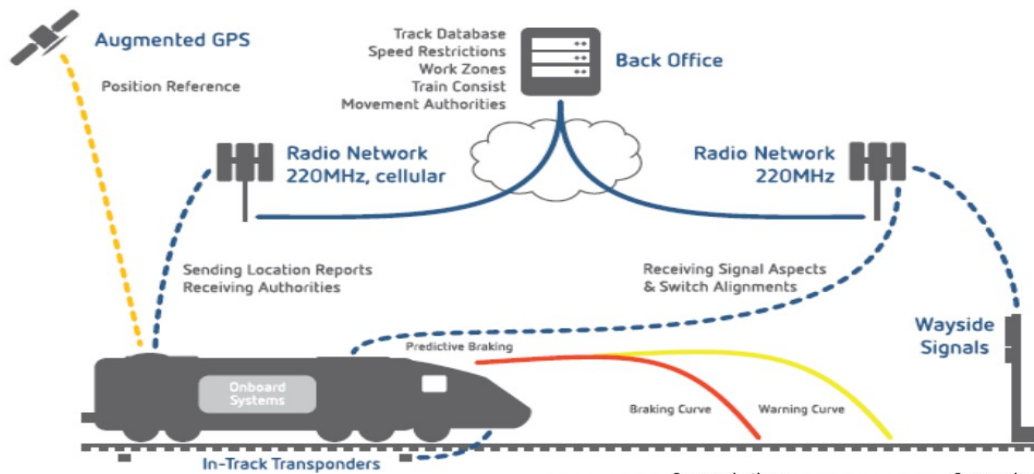


Fig. 2 PTC operation [1].

GPS is intended to mitigate the significant cost of deployment of in-track transponders (balises) on railroads nationwide (Joint Council on Transit Wireless Communications 2012). The solution can be enhanced by track beacons or differential GPS stations hybridized with wheel sensors [5], demonstrates the capability of low-cost on-board components to discriminate cross-track with data fusion without any trackside infrastructure.

European Train Control System (ETCS), [6] GNSS can provide the absolute positioning function in ETCS and hence, can be a drop-in replacement to the system of balise and the onboard BTM. This is shown on the right of Figure 3. This development has global implications as ETCS is used worldwide. For example, the Chinese Train Control System (CTCS), versions 2 and 3, used on highspeed rail lines, are also based on ETCS. However, for GNSS to be used, it needs to meet or exceed the performance of the current balise system.

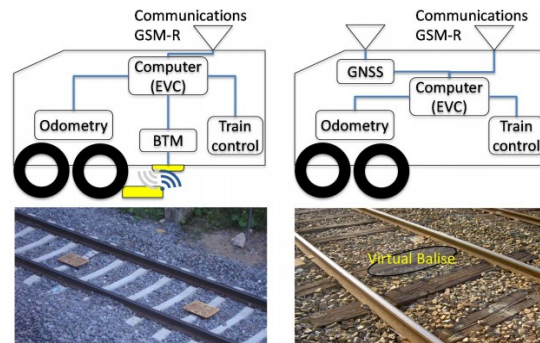


Fig.3 ETCS with Balise Transmission Module (BTM), European Vital Computer (EVC), Odometry and GSM-R (rail) communications (left) with balises (in yellow, picture on bottom) and ETCS concept with GNSS drop in replacement to balise system utilizing virtually defined balise based on GNSS positions.

Unmanned railway crossing alarm and train detectors [7], the unmanned railway crossing alarm system alarms crossing when a train approaches. The mechanism involves automatic train detection without any human intervention. The train detection sent over RF wireless transmission to level crossing receiver. Alert sound and light alarm flashes. This, therefore, alarms pedestrians and vehicles cross the track. This solution avoids accidents that occur at the unmanned railway crossing. It functions 24×7.



Fig.4 Unmanned Railway Crossing Alarm - FORBIX SEMICON

Following is how the system works:

Step1: Incoming train detected (2 to 3 Kilometers) before the level crossing

Step2: Train detected signal is sent over long distance wireless transmitter and receiver

Step3: When a train approaches, a siren blows at the level crossing. This associate with alert light flashing at the crossing

Step4: Once the train departs, the unmanned crossing, the siren/alarm stops

Unmanned level crossing gate warning system (ULCGWS) [8][9] is a microprocessor based system having dual GSM, GPS and RF modules. The device broadcasts its location data for local devices within a 2000m range. When a train passes through any location, data gets exchanged between the train and the LC gate. To detect and convey the arrival of the train at the LC gate, and to provide a backup to this RF communication-link between the train and LC gate, the system uses load cells located approximately 2km from the LC gate in both the directions.

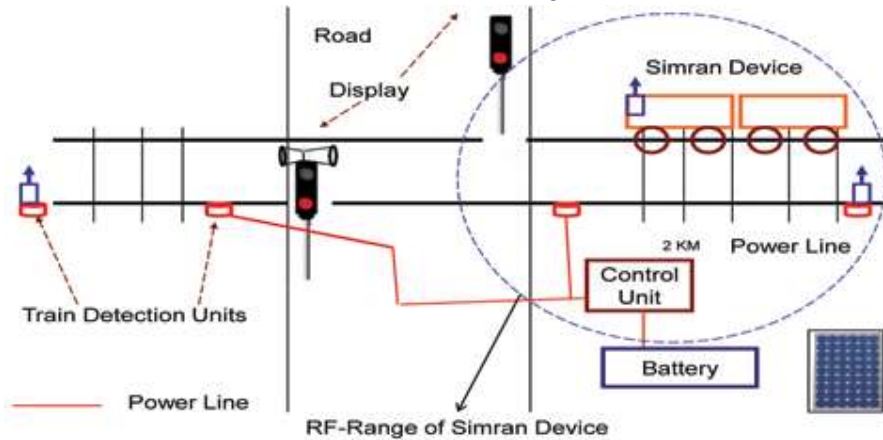


Fig. 5 Automatic railway crossing warning system ULCGWS on base RF and GPS

When train nears the LC gate, information about its arrival is passed on to the LC system. The LC gate equipment actuates a red signal and an audible warning using 200db hooters placed both sides of the road. The system uses a three aspect colour light signal, with the red and yellow lights used for the warning signals and the hooters being used in place of the third colour light. The system stops the warning lights and audio warnings once the train passes the level crossing.

In the country: in the face of the potential for railway traffic accidents that are always lurking daily. The Ministries and sectors have also promoted the application of scientific research to renovate the railway system, which is too outdated, in order to localize domestic products, take initiative in technology, enhance safety assurance and throughput capacity of the railway industry. Regarding the technology of the research direction, there have been many applied research works, proving the ability to work reliably and effectively, such as, against falling asleep devices, speed recording devices train [10] (Unit in charge of implementation: University of Transport and communication), signal equipment stern train, railway crossing monitoring and control equipment [11][12] (Unit in charge of implementation: Institute of Transport Science and Technology), automatic train protection system (I-ATP) [13] (Unit in charge of implementation: national center for technological progress), and [14-16].

The systems applying radio communication solutions and GPS technology researched and applied

at local and foreign have proven their ability to work reliably, safely and with high availability, great effected for the railway industry. However, for the railway crossing warning system applying radio communication solutions and GPS technology, there is no application product or published research work in the country. In the world, a number of countries have researched and applied it, which has brought about high economic efficiency, safety and convenience in the management, operation. However, from practice, to achieve even higher efficiency, it is necessary to add more functions to the system, such as: need to transmit the image of the front railway crossing to the train to help drive train easy to accurately observe the smooth state at the railway crossing (as well as save information so that it can be easily traced when an incident occurs, improving the train driver). It is also important to centrally manage and monitor the equipment at the railway crossing and the equipment on the locomotive in real time, in order to promptly detect and handle problems when there is a problem, to avoid the maximum the delay in operation as well as other risks that may arise during the operation.

3. System design research

3.1. Design scheme and operating principle of the system

From the analysis, evaluation and through actual research, the author proposes to build a systematic overview scheme, also sets out the principles and

capabilities of the system to be able to solve these problems exist. The system operation on the basis of radio communication, microprocessor and global

positioning system. The system overview scheme is as follows:

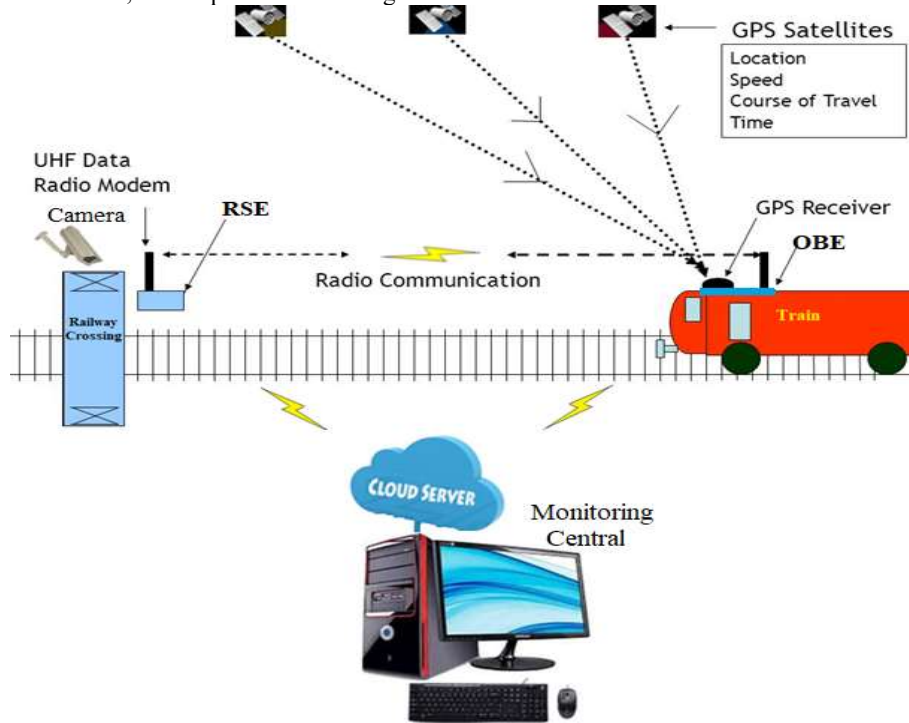


Fig. 6 The system overview scheme of ARCWS on base RF and GPS

The main components of the system:

- 1- Equipment on the locomotive: Including GPS module; RF module; information display HMI, images at the railway crossing; The memory stores a database of the coordinates of the railway crossing, the corresponding speed instructions for each segment railway and the central processor.
- 2- Equipment at the railway crossing: Including RF module, horn, lights, central processing unit and system to monitor the working status of the system.
- 3- Monitoring center: allows real-time monitoring of equipment at the railway crossing and on the locomotive.

The operating principle of the system: Basically the system operation as follows, the device on the locomotive continuously updates data from the satellite about the position and speed of the train. Analyzing the actual speed compared to the permitted speed command at that section, if the threshold is exceeded, the device on the locomotive will warn the train driver, and record it in memory (serve investigation, remind the driver). Compare the train's position with position of the railway crossing (stored in the locomotives's memory), when approaching the railway crossing at a certain distance (this warning distance can be set flexible to suit the characteristics of each railway crossing) at this time, the system warns the train driver that the train is coming to the railway crossing, and at the same time sends information to the device at the

railway crossing to warn the horn/light at the area. When the train passes through the railway crossing area, the system automatically cancels the active mode of the horn/light to interrupt the warning. In addition, during the work, the system always checks and measures the working parameters of the controller and peripheral devices to be able to determine the working status of the system, to be timely provides the safest operating mode, the device gives warnings on the spot as well as sends alarm information to the server and supervisor for promptly handle it when the problem occurs.

The system has two most important components, the radio communication function between the equipment on the train and the equipment at the railway crossing and the acquisition of data from the satellite. For radio communication capabilities, a module capable of communicating reliably even when covered (by buildings, mountains and forests). However, for data collection from satellites, there are some areas where the device will lose signal (when the train enters the tunnel), the processing algorithm is performed as follows: determine the adjacent coordinates of areas that are likely to lose satellite signals, when the train enters that area, it will now be based on the direction of movement, train speed and corresponding time to determine the position and calculate distance between the train and the railway crossing ahead.

Communication solution between equipment at the railway crossing and equipment on the locomotive to the center: this is the task to monitor operating status of the system, so that can be used GSM transmission, making the design simple and low cost while still ensuring the requirements of the system. For the transmission of data information between the equipment at the crossing and the equipment on the locomotive, this is an important task not only related to monitoring but also related to the control of warning devices at the railway crossing. The railway crossing, so it is necessary to ensure reliability, availability and fast response, the chosen solution is the application of Lora technology, this is a new, advanced technology and is increasingly being applied in many areas world. LoRa (Long Range) is a modulation technique based on Spread-Spectrum technique which is a variant of Chirp Spread Spectrum (CSS) that maintains the same low power characteristics as FSK modulation but significantly increases communication range, which helps to transmit significantly longer distances than other techniques. Lora technique uses chirp frequency with a linear transformation of frequency over time to encode information, this technique makes the frequency difference between transmitter and receiver easily eliminated in the decoder code and also helps to be immune to the Doppler effect (this is an important factor, applied to equipment moving at high speed). The frequency difference between the transmitter and the receiver can be up to 20% of the bandwidth without affecting the decoding performance.

To determine the position and speed of the train in real time base on GPS technology, using the module of U-blox, this company has launched the first GNSS module based on the technology blox F9 with centimeter level accuracy. The ZED-F9P is the first multichannel receiver to use GNSS signals from all four GNSS satellites simultaneously (GPS, GLONASS, Galileo and BeiDou). Combining GNSS signals from multiple bands (L1/L2/L5) and RTK technology allows the ZED-F9P to achieve centimeter-level accuracy and can receive more signals from satellites anywhere any time, allowing for optimized accuracy even in challenging environments such as cities.



Fig.7 Module GNSS u-blox M8



(a)



(b)

Fig.8 Equipment circuit on the locomotive: (a) display the allowed speed at the road; (b) notice to the train driver through loudspeaker

Power supply for warning system at railway crossing using solar power and battery backup. In order to bring about high efficiency in power source stability, simplicity in construction, and at the same time saving labor to maintain the system and being less affected by external influences (due to solar power source is an independent source, there is no need to dig holes for cables to go from the grid to the system, the repair of roads - infrastructure, power grid, etc. will not affect the system).



Fig. 8 system model using solar power at the railway crossing

3.2. Safety principles (fail safe)

For the railway industry, safety comes first. In order to meet that requirement, in addition to factors such as component quality, design, etc., the system needs to achieve key objectives such as availability, reliability, and failure in safety, ... [17]. To achieve the standards in this regard, the system is basically establish as follows: Depending on the characteristics of each railway crossing, the system will issue a warning horn/light at the railway crossing when when the train distance there from 2km to 3km (calculated according to the closing time from [18]). To ensure safety, the radio transceiver module selected is the transceiver with a distance of > 3.5km. When the train approaches the railway crossing area, the transceiver module on the locomotive and the transceiver module at the RC will receive signals and communicate continuously with each other to accurately determine information from both sides (in an area with many railway crossing close together, the device on the locomotive can be connected to many devices at the same time, for each device at the RC there is a different code, based on the location and the direction of movement of the train so that it is easy to determine the forward RC closest to the train). In case the GPS module in the device on the locomotive is faulty, the device on the locomotive will warn the train driver, and at the same time send information to the device at the crossing to give a warning of the horn/light at the RC in area. During the train is moving, the device on the locomotive continuously collects satellite data on position and speed, comparing it with the position of the RC (stored in memory) when the train approaches the RC, it is necessary to give a warning, but the RF transceiver module at RC or on the locomotive fails, resulting in the equipment on the train and the horizontal line not connecting to each other, the equipment on the locomotive will warn the train driver to provide a

mechanism to reduce the train speed to a minimum or brake the train. The equipment at the RC and the device on the locomotive continuously check the operation of the controller and peripheral devices, when there is any error, the system automatically returns the system to operation mode fail-safe, and at the same time give warnings on the spot as well as send information to the server and manager for early detection and timely troubleshooting. In addition, regarding radio transceivers: The system needs to have reliability in terms of security in transmitting and receiving information: The solution given on the principle of selecting hardware and software requires encryption and Decrypt information, control data using CRC-32 (32-bit) checksum algorithm.

Analysis of the affect of doppler effect: Radio communication will be affected by the doppler effect [19] when the transmitter or receiver is moving. In the automatic railway crossing warning system, communication between the equipment at the RC and the equipment on the locomotive, the train is moving, the equipment at the RC is stationary.

With:

f_0 : frequency when sending

f : frequency when receiving

v : train speed

c : speed of signal wave (say 300,000 m/s)

Suppose, communication on the sending frequency is 433 MHz ($f_0 = 433$), when the train is moving at 160 km/h (~45 m/s), the frequency when receiving f is changed by the doppler effect in terms of [19] is:

When device on the locomotive sends a message to the device at the RC:

$$f = f_0 \left(\frac{1}{1+v/c} \right) \approx 432,935 \text{ (MHz)}$$

When the device at the RC sends a message to device on the locomotive:

$$f = f_0 \left(1 + \frac{v}{c} \right) \approx 433,065 \text{ (MHz)}$$

The radio communication technology used in the RC warning system is Lorawan spread spectrum wideband. The Lorawan communication devices of the system are divided into 32 channels, each channel has a bandwidth of 1 MHz (from 410 - 441 MHz). That is, assuming channel number 23 (nominal frequency 433 MHz) will accept frequencies in the range 432.5 – 433.5 MHz. Thus, according to the above calculation results, we can see that the frequency shift by the doppler effect is not different beyond the operating frequency range for Lorawan. Several reference trials, [20] have concluded that Lorawan is immune to the doppler effect.

3.3. System scalability

Currently, on the locomotives of Vietnam railways, there are quite a lot of monitoring and supporting devices for train drivers such as: speed measurement device, against falling asleep device, stern train signal

device, ... with integrated GPS system and RF radio communication solution. Therefore, it is completely possible to integrate these component devices with the automatic railway crossing warning system, to create a equipment with enough features for automatic monitoring, driving assistance, warning alerts to train drivers and warning at RC to reduce investment costs, ease maintenance and repair, compact equipment and reduce system complexity.

The system can integrate more cameras at the RC area, the image at the intersection will be transmitted automatically and displayed on the screen on the locomotive. When the train approaches the RC area, the radio transceiver module between the equipment on the locomotive and the RC will be connected together, at this time the image at the RC from the camera is continuously sent to the display on the locomotive, helping train drivers grasp the actual situation at the RC and promptly respond when there is a problem with the RC.

Integrating with the incident detection system at the RC: In the case of the RC area there is a problem due to the vehicle stalling, malfunctioning, etc., leading to being stuck at the RC. Equipment at the RC such as camera-image processing systems, radar scanners, lidar, etc. will detect obstacles at the RC, thereby continuously broadcasting warning messages. When the train approaches the RC area, the device on the locomotive will receive information to warn the train driver through a siren and image for the train driver to make a decision timely to avoid the consequences of traffic accidents at the RC.

4. Conclusion

On the basis of theoretical design analysis, the actual testing process also gives good results. The ability to collect satellite data and ensure communication, not affected by interference by external influences. Communication distance can be achieved up to 4km in inner city and 7km in suburban area with train speed over 70km/h



Fig.9 Test the system's ability to work in actual operating conditions

Within the framework of the article, the design as well as the working ability of the automatic railway crossing

warning system based on the RF solution and the GPS system technology. This is the premise to develop standards, conduct trial production to perfect products for installation and practical applications. Contributing to reducing railway traffic accidents, minimizing investment costs, facilitating the installation process for areas where there is no automatic RC warning system and gradually replacing foreign products. In addition, helping to master the technology, facilitate the installation, operation and maintenance of the system.

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